

VENU R

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EDUCATION

Sir M Visvesvaraya Institute of Technology, B.E. in Information Science

Sept 2023 – Jul 2027

GPA: 8.61

TECHNICAL SKILLS

Languages: Java, C, JavaScript, SQL

Frameworks & technologies: React, FastAPI, Node.js, Express.js, Flask, HTML, CSS

Databases & Tools: MySQL, PostgreSQL, Git, Kali Linux

Concepts: DSA, OOP, DBMS, OS, Multithreading, Computer Networks, SDLC, REST APIs

PROJECTS

Enterprise Task & Workflow Management System

GitHub, Demo

- Built a full-stack web platform for team collaboration and project tracking with role-based access for Admin, Manager and Employee users
- Implemented JWT based authentication and RBAC to enforce role-specific authorization across application modules.
- Integrated Socket.IO for real time task notifications and dashboard updates supporting concurrent users.
- Developed REST-based backend services and integrated database operations for managing users, tasks and project workflows.
- Containerized and deployed the backend on Azure Container Apps and deployed the frontend on Vercel.

Tech Stack: MERN, Socket.IO, JWT, bcrypt, Vercel, Docker, Azure Container Apps

AttentionPay – Transaction fraud detection

GitHub, Demo

- Developing a two-stage security platform combining BERT based phishing detection with TabTransformer based transaction fraud detection.
- Built a FastAPI backend with URL analysis, API route integrity and impossible travel checks to identify suspicious activity before transaction processing.
- Implemented transaction feature engineering, model inference and explainability components with PostgreSQL for transaction data management.
- Developing a React based dashboard to present phishing and transaction risk predictions with interpretable results.

Tech Stack: FastAPI, React, PostgreSQL, PyTorch, Transformers, TabTransformer

Cyber Threat Detection System using IoT

GitHub

- Developed a 4 node ESP32-based IoT security monitoring system to collect real time sensor data and network activity logs in a simulated environment.
- Simulated DoS, spoofing, burst and replay attacks using Wireshark, Nmap and hping3 to generate labelled network traffic data.
- Built a Flask backend with a scikit-learn anomaly detection model to identify abnormal network activity and a dashboard to monitor security alerts.
- Generated and analysed 1,000+ network traffic records for anomaly-detection model training and evaluation.

Tech Stack: Flask, scikit-learn, ESP32, Arduino, Wireshark, Nmap, hping3, Kali Linux

RiskLens — AI-Powered Risk Assessment Platform

GitHub

- Developed a full-stack risk assessment platform for evaluating operational, financial and technical project risks.
- Trained an ML model on historical project data to predict risk likelihood and impact across multiple project parameters.
- Built a Flask backend and React dashboard to provide risk scores, severity analysis and visual risk insights.

Tech Stack: Flask, scikit-learn, Pandas, NumPy, MySQL, React, JavaScript

CERTIFICATIONS

Cybersecurity and Privacy - NPTEL, 2024

Cryptography and Network Security - NPTEL, 2025